

Reputation-Aware Truth Discovery for Robust Evidence Fusion in RAG Pipelines

Diogo Ribeiro^{1,2}[0009–0005–0713–5842] and João Marco Silva^{1,2}[0000–0003–4621–5288]

¹ INESC TEC, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal
{diogo.p.ribeiro, joao.marco}@inesctec.pt

² University of Minho, Rua da Universidade, 4710-057 Braga, Portugal
{d13847@uminho.pt, joao.marco@di.uminho.pt}

Abstract. Retrieval-Augmented Generation (RAG) improves Large Language Models (LLMs) by grounding responses in external evidence, but the reliability of retrieved information often varies across heterogeneous sources, leading to conflicting evidence and reduced trustworthiness. This challenge is particularly acute in information-sparse settings, where robust decision-making depends on principled mechanisms for aggregating uncertain and potentially unreliable inputs. This paper presents Skald, a reputation-aware truth discovery method for robust evidence fusion in RAG pipelines. Skald combines a truth discovery module for heterogeneous data with a reputation mechanism that preserves and updates source reliability across executions, enabling dependable sources to exert greater influence over time while penalizing unreliable ones. Unlike conventional truth discovery approaches that reinitialize source reliability at each run, Skald leverages historical source behavior to improve adaptability in dynamic multi-source environments. Evaluations on real-world datasets show that Skald consistently outperforms established methods such as TruthFinder and CRH across multiple similarity metrics. These results indicate that integrating reputation-aware truth discovery into RAG-oriented evidence fusion provides a principled foundation for improving robustness, trustworthiness, and decision quality in intelligent systems operating under uncertainty.

Keywords: Retrieval-Augmented Generation · Data Fusion · Truth Discovery · Reputation Systems · Uncertainty Handling.

1 Introduction

Retrieval-Augmented Generation (RAG) integrates information retrieval with generative language models, whereby, given a user query, the system retrieves relevant passages from an external corpus and conditions Large Language Models (LLMs) on both query and evidence to produce a grounded answer [11,20]. This approach taps into up-to-date factual knowledge beyond parametric memory, mitigating hallucination and enhancing accuracy [20,31]. However, retrieved content may vary in reliability, and conflicting or low-quality information can

mislead the generator, resulting in erroneous or misleading content [2,15,27]. Addressing this requires mechanisms to assess, weight, or filter retrieved evidence by trustworthiness.

Truth Discovery (TD) provides a systematic framework for reconciling conflicting information from multiple sources by estimating the reliability of each contributor and the credibility of the information they provide. Algorithms such as TruthFinder [26] and CRH [12] iteratively refine estimations of data veracity based on source agreement, progressively elevating information backed by dependable sources. In contrast to supervised Machine Learning (ML) and LLMs, which require extensive labeled datasets and may hallucinate in low-resource or specialized domains [5,9], TD operates without predefined ground truths, instead relying on source reliability. Nevertheless, conventional TD methods require multiple iterations to converge and fail to track source reliability over time, resetting uniform source weights at each execution [13,21].

This work proposes Skald, a novel solution that fuses TD with a reputation system to mitigate the impact of unreliable or conflicting retrieved evidence in RAG pipelines. In this setting, retrieved information constitutes heterogeneous evidence that must be reconciled under uncertainty, requiring mechanisms that account not only for agreement between sources but also for their trustworthiness over time. Skald is designed as a modular evidence-fusion layer for RAG pipelines, enabling its integration between retrieval and generation to assess, consolidate, and weight competing evidence before answer construction. By embedding reputation scoring directly into the TD module, Skald accumulates reputation for each source based on the quality of its contributions. Unlike existing approaches, Skald retains and leverages historical reliability information when computing source influence, thereby enabling dependable sources to exert greater impact in subsequent executions while penalizing unreliable ones. Through this integration, Skald supports more robust evidence fusion and enhances the reliability of downstream intelligent decision-making in dynamic, multi-source environments.

2 Background on Truth Discovery

TD is a data fusion approach aimed at resolving conflicts in information provided by multiple sources by identifying the values that most accurately represent reality [1,3]. Some of the most well-known algorithms in literature include TruthFinder [26], 2-Estimates [4], 3-Estimates [4], Investment [18], LTM [30], GTM [29], LCA [19], and CRH [12]. These algorithms employ different strategies for TD, with a key distinction being the types of data they support. Among them, only TruthFinder and CRH provide mechanisms for handling heterogeneous data [13], *i.e.*, a combination of categorical, string, and continuous data [1].

Machine learning-based methods [24,28] require large volumes of training data, which may not always be available [10]. The lack of training data is particularly problematic in RAG scenarios, where the reliability of retrieved documents

can vary widely and may not be known a priori, thus limiting the applicability of ML approaches.

TruthFinder employs Bayesian analysis to iteratively estimate source reliability and infer truths [26,13]. CRH addresses truth inference by casting it as an optimization problem that minimizes the weighted deviation between observed and inferred values [12]. Both approaches compute source reliability at the end of executions. However, these estimates are not preserved for subsequent runs, leading to uniform source weighting. This limitation, often referred to as the source reliability initialization problem, remains an open issue in the literature [21,13].

More recent approaches [14,25,32] introduce dynamic TD methods for data streams, enabling continuous updates of both inferred truths and source reliability. While effective in adapting to evolving information, these methods do not support heterogeneous data. Furthermore, the approaches in [14] and [25] do not retain historical data, which limits their ability to revise past inferences when sources change their behavior or are later found to have provided inaccurate values.

3 Proposed Approach

As illustrated in Figure 1, Skald is organized into three interacting components: the TD module, which infers the most credible values from heterogeneous evidence; the evaluation module, which converts the resulting confidence scores into ratings for each source contribution; and the reputation module, which tracks and updates source trustworthiness over time. In this architecture, evidence and source information are provided as input to the TD module, which produces consolidated evidence together with confidence scores. These scores are then used by the evaluation module to derive ratings, which are incorporated by the reputation module to maintain reputation data across executions. This interaction enables Skald to combine evidence consolidation with persistent source reliability modeling, thereby supporting more robust decision-making in dynamic multi-source environments.

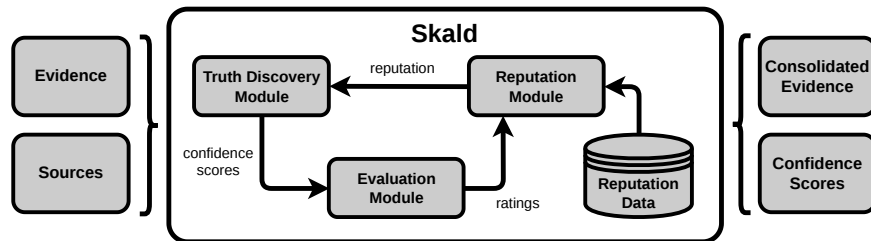


Fig. 1: Overview of Skald’s components and their interactions.

The TD module processes a set of values and outputs the one most likely to be true, along with confidence scores for each input. TruthFinder [26] was chosen as the starting point because, among the algorithms in Section 2, it uniquely supports heterogeneous data while providing well-defined confidence scores in the $[0, 1]$ range [13]. We diverge from the methodology proposed in TruthFinder by presenting each source’s reputation score $r(s)$ at the outset of the TD process. This approach eliminates the need for multiple successive iterations, since only the confidence score for each value is required to be calculated.

$$\omega(v) = \sum_{s \in \mathcal{S}(v)} -\ln(1 - r(s)) \quad (1)$$

$$\sigma(v) = \sum_{i_{v'}=i_v} \omega(v') \cdot \text{imp}(v' \rightarrow v) \quad (2)$$

$$\phi(v) = \frac{1}{1 + e^{-\gamma \cdot (\omega(v) + \rho \cdot \sigma(v))}} \quad (3)$$

The first step computes each value’s weight $\omega(v)$ via Eq. 1, based on the reputation r of sources providing value v (denoted by $\mathcal{S}(v)$). To account for alternative values and their relationships, we define the implication metric $\text{imp}(v' \rightarrow v)$, ranging from -1 (contradiction) to 1 (high similarity). For categorical data, direct comparison yields 1 for matches and -1 for contradictions. For strings, Jaro–Winkler similarity [22,23] handles typos and phonetic variants; for continuous data, Euclidean distance is used. All scores are normalized to $[-1, 1]$, and the implication functions can be extended to support additional data types.

The influence of remaining information about a subject, $\sigma(v)$, is computed via Eq. 2, integrating implication between values. A positive $\sigma(v)$ indicates support from other values, whereas a negative $\sigma(v)$ denotes contradiction.

Calculating the confidence score $\phi(v)$ from $\omega(v)$ and $\sigma(v)$ presents three challenges. First, source dependence (e.g., copying) can inflate scores. Therefore, a dampening factor $\gamma \in [0, 1]$ is needed to limit score evolution. Second, the influence of related information needs to be controlled by $\rho \in [0, 1]$, where $\rho = 0$ excludes $\sigma(v)$ and $\rho = 1$ fully applies it. Third, contradictions with higher-reputation sources may yield negative scores. Modeling $\phi(v)$ with a sigmoid-like function [8] in Eq. 3 ensures that $\phi(v) \in [0, 1]$ even if $\gamma(\omega(v) + \rho \sigma(v))$ lies outside this interval by a significant margin.

The evaluation module converts confidence scores $\phi(v)$ into source ratings $\mathbf{r}_s$, defined as $\mathbf{r}_s = (r_1, r_2, \dots, r_k)$, $r_{\text{level}} \in \{0, 1\}$, $\text{level} = 1 \dots k$. The index of the level set to 1 is determined by $\lfloor \phi(v) \times k \rfloor$, where k represents the size of $\mathbf{r}_s$, ensuring an integer index.

The reputation module updates source reputation scores $r_s \in [0, 1]$ using received ratings $\mathbf{r}_s$. Contributions are mapped into a Bayesian binary state space, and reputation scores are obtained by updating *beta* distributions. Since the state space exceeds two levels, a Dirichlet distribution is used [7,16]. The parameter k specifies the number of rating levels, with higher k yielding finer granularity at the cost of computational overhead. A longevity factor $\lambda \in [0, 1]$ defines the

weight of recent ratings, controlling rating aging: $\lambda = 0$ forgets ratings after one update, while $\lambda = 1$ preserves all past ratings.

The cumulative rating vector for source s is $\mathbf{R}_s = (R_s(i) \mid i = 1, \dots, k)$, which is updated by adding the new rating $\mathbf{r}_s$ to the previous aggregate as $\mathbf{R}_{s,iter+1} = \lambda \cdot \mathbf{R}_{s,iter} + \mathbf{r}_s$. The multinomial probability vector $\mathbf{S}$ (Eq. 4) incorporates the *a priori* constant C (typically 2) to moderate new observations' effect in the Dirichlet distribution, with base rate $a_k = [1/k]^k$ under uniform assumptions. If no ratings exist, $\mathbf{R}_s = [0]^k$ and $\mathbf{S} = [1/k]^k$. The multinomial probability density function requires $\sum_{i=1}^k a(i) = 1$ and $\sum_{i=1}^k S(i) = 1$, where i indexes rating levels.

$$\mathbf{S} : \left(\frac{R_s(i) + Ca(i)}{C + \sum_{i=1}^k R_s(i)} \mid i = 1 \dots k \right) \quad (4)$$

While the multinomial probability representation provides detailed information, it requires substantial space when k is large. A more compact approach represents the reputation score as a single value in a predefined interval (*e.g.*, $[0, 1]$). This is achieved by assigning a point value to each rating level i and computing the normalized weighted point estimate in Eq. 5. With k rating levels evenly distributed over $[0, 1]$, the point value for each level is $\frac{i-1}{k-1}$. For a source with no ratings, Eq. 5 yields 0.5, the initial value of r_s in the system.

$$r_s = \sum_{i=1}^k \frac{i-1}{k-1} S(i) \quad (5)$$

Within a deployed RAG pipeline, Skald is conceived as a modular evidence-fusion layer positioned between retrieval and generation, as illustrated in Figure 2. Given a user query, the retriever accesses the knowledge base and returns a set of retrieved evidence items associated with potentially different sources, as well as the candidate values or claims that can be derived from them. As shown in the figure, these retrieved elements constitute the input to Skald, which operates over heterogeneous and potentially conflicting information rather than directly over raw user queries. On this basis, Skald performs TD to estimate the most credible values, computes confidence scores for competing evidence, and updates source reputations across executions through its reputation mechanism. The output of this process is a consolidated and reliability-aware representation of the retrieved evidence, denoted in Figure 2 as consolidated evidence, which is then passed to the generator to support answer construction. In this way, Skald acts as an intermediate trust and aggregation layer that filters, weights, and reconciles retrieved information before generation, thereby improving the robustness and trustworthiness of downstream responses.

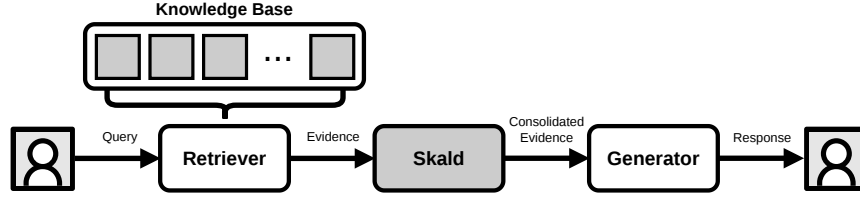


Fig. 2: Overview of Skald as an evidence-fusion layer within a RAG pipeline.

4 Proof of Concept

This section evaluates a publicly available implementation of Skald¹ at the evidence-fusion stage of the pipeline, isolating the effect of reputation-aware TD from the variability introduced by retrieval and generation stages. This experimental design enables a controlled assessment of Skald’s ability to reconcile conflicting information, estimate source reliability, and identify the most credible values across heterogeneous data. Accordingly, the objective of this evaluation is not to measure end-to-end answer quality in a complete RAG system, but rather to examine the specific contribution of Skald as the intermediate layer that consolidates and weights evidence prior to generation. In doing so, the analysis provides a principled foundation for subsequent end-to-end integration in deployed RAG pipelines.

All experiments were performed on a Dell OptiPlex 5000 with an Intel i5-12500 processor, 256GB NVMe storage, and 64GB memory, running Rocky Linux 9.4. All baseline methods, as well as Skald, are implemented using Python 3.12.3. Skald’s accuracy is assessed on three publicly available datasets commonly used in the literature² [26,3,19]: *books*, *flights*, and *stocks* datasets. Each includes a gold standard representing ground truth for accuracy evaluation.

Results are compared against two baselines, TruthFinder and CRH, chosen not only for their established status in the literature, but also because they are among the most appropriate points of comparison for Skald, given their support for heterogeneous data via suitable implication functions. TruthFinder is of particular interest because it serves as the methodological basis for Skald’s TD module, while CRH provides a strong alternative formulation for truth inference over heterogeneous inputs.

Accuracy is measured by computing the normalized similarity between each method’s output and the gold standard using multiple metrics: Jaro–Winkler (JW), Sørensen–Dice (SD), and Damerau–Levenshtein (DL) for strings [22,23]; Euclidean and Manhattan distances for continuous data [17]. Execution time and accuracy are averaged over 10 runs, with Skald’s input order randomized before each run. The parameter k is set to 100 to prevent sources with a high number of

¹ <https://github.com/dissect-project/skald>

² Full datasets available at <http://lunadong.com/fusionDataSets.htm>.

Table 1: Accuracy results (in %) of Skald compared to TruthFinder and CRH

Algorithm	Books			Flights		Stocks	
	JW	SD	DL	Euclidean	Manhattan	Euclidean	Manhattan
TruthFinder	88.35%	76.18%	62.32%	99.99%	99.99%	97.06%	97.06%
CRH	91.24%	84.57%	71.68%	99.99%	99.99%	91.83%	91.83%
Skald	94.15%	90.30%	80.32%	99.99%	99.99%	99.18%	99.18%

contributions from attaining disproportionately high reputation scores despite having a low number of ratings.

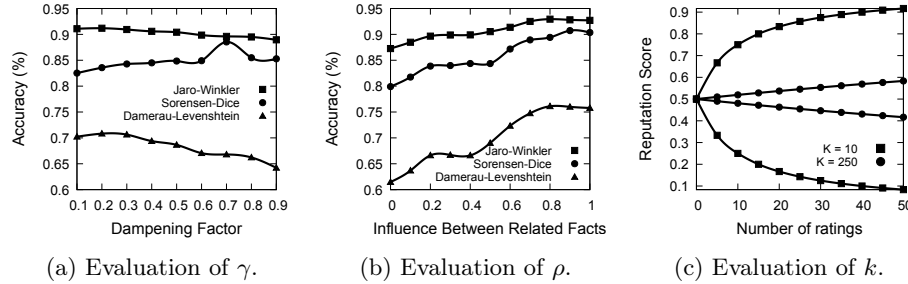
Table 1 highlights the accuracy results of Skald compared to the baseline methods. Skald outperforms or matches both TruthFinder and CRH across all datasets and similarity metrics, surpassing them on average by 4.57% and 5.89%, respectively. Notably, in the *books* dataset, Skald achieves an average accuracy of 88.26%, outperforming TruthFinder by 5.76% and CRH by 12.64%. Similar trends are observed with the *stocks* dataset, where Skald achieves an average accuracy of 99.18%, surpassing TruthFinder by 2.12% and CRH by 7.35%. For the *flights* dataset, all three methods achieve similarly high average accuracy on continuous data, reflecting the dataset’s minor deviations. These results demonstrate that Skald can accurately identify the value most likely to represent the truth in both high and low-variability scenarios, highlighting its adaptability.

This evaluation indicates that Skald consistently outperforms established TD algorithms across datasets of varying sizes, from the *books* dataset (with 33,971 values) to the *stocks* dataset (with 12,137,940 values). Notably, the 2.12% and 7.35% accuracy gains over TruthFinder and CRH on the *stocks* dataset, equate to approximately 257,000 and 892,000 more correctly identified values, underscoring the practical significance of these improvements. These gains confirm that Skald’s approach substantially enhances truth identification reliability, thereby reinforcing its suitability as an evidence-fusion mechanism for RAG pipelines.

An analysis into Skald’s adjustable settings was carried out using the *books* dataset, with each result averaged across five runs. The first assessed parameter is the dampening factor γ , with $\rho = 0.5$ and $k = 10$. Tests incremented γ by 0.1 from 0.1 to 0.9. Figure 3a shows that varying γ induces accuracy fluctuations of about 5% for SD and DL, and approximately 2% for JW.

The second parameter evaluated is the influence between related facts ρ , which adjusts how related facts affect the confidence score (Eq. 3). Tests varied ρ from 0 to 1 in 0.1 increments. Figure 3b shows that accuracy increases with ρ , peaking around 0.8–0.9. Higher ρ (e.g., above 0.5) leverages supportive and contradictory relationships between values, thereby improving accuracy.

The final tested parameter is k . Adjusting k alters the dispersion of source reputation scores, since it determines the number of ratings required for updates via the Dirichlet distribution [6]. Lower values of k yield more widely spread scores in $[0, 1]$, with high-reputation sources near 1 and low-reputation sources near 0, whereas higher values (e.g., 250) concentrate scores around the initial value of 0.5, thus controlling the rate of reputation evolution. Figure 3c illustrates

Fig. 3: Evaluation for different γ , ρ , and k values.

reputation trajectories for $k = 10$ and $k = 250$ under consistently best and worst ratings, demonstrating how k tunes the evolution rate. When sources provide few contributions, a lower k enables quicker reputation updates, whereas a higher k stabilizes scores for frequently contributing sources by requiring more ratings for substantial changes.

5 Conclusion

In this paper, we introduced Skald, a reputation-aware TD approach for robust evidence fusion in RAG pipelines. By combining TD over heterogeneous evidence with persistent source reputation modeling, Skald provides a modular evidence-fusion layer that consolidates conflicting information before generation. Experimental results on real-world datasets show that Skald consistently outperforms established TD methods across multiple similarity metrics, demonstrating improved reliability in identifying the most credible values. These findings support the use of persistent source reliability as a principled mechanism for improving evidence consolidation in dynamic multi-source environments.

The results indicate that reputation-aware evidence fusion can strengthen the robustness of RAG pipelines, particularly in information-sparse settings where limited, noisy, or conflicting evidence increases the importance of trust-aware aggregation. While the present evaluation focuses on the evidence-fusion component in isolation rather than on end-to-end generation quality, this design enables the contribution of Skald to be assessed independently of retrieval and generation variability and provides a principled basis for future integration into deployed RAG workflows. A current limitation of this study lies in the evaluation being conducted on structured truth-discovery benchmarks rather than directly on retrieved unstructured evidence within a complete RAG system.

A first direction for future work resides in performing an end-to-end evaluation within complete RAG systems, including the measurement of downstream effects on factual accuracy, answer consistency, robustness to misleading evidence, and sensitivity to retrieval quality. A second direction is the extension of Skald beyond structured and semi-structured inputs and toward unstructured textual evidence, where claim extraction, evidence alignment, and contradic-

tion detection would need to be incorporated more explicitly into the pipeline. A third direction concerns richer reliability modeling, including topic-sensitive and context-dependent reputation mechanisms that can capture variations in source trustworthiness across domains, tasks, or temporal conditions. Together, these directions would allow Skald to evolve from an evidence-fusion component validated in isolation into a broader trust-aware mechanism for improving the reliability of end-to-end RAG systems.

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